

1. Assign roles (admin, ticket clerk) and control access to sensitive data.

-- Create user

```
CREATE USER 'admin_user'@'localhost' IDENTIFIED BY 'Admin@123';
```

```
CREATE USER 'clerk_user'@'localhost' IDENTIFIED BY 'Clerk@456';
```

-- Admin: has all authorities

```
GRANT ALL PRIVILEGES ON CinemaDB.* TO 'admin_user'@'localhost';
```

-- Clerk: just can view + sell tickets

```
GRANT SELECT ON CinemaDB.MOVIES TO 'clerk_user'@'localhost';
```

```
GRANT SELECT ON CinemaDB.SCREENINGS TO 'clerk_user'@'localhost';
```

```
GRANT SELECT, INSERT ON CinemaDB.TICKETS TO 'clerk_user'@'localhost';
```

TEST CASE

-- Log in by clerk_user (using cmd/bash): cd "C:\Program Files\MySQL\MySQL Server 8.4\bin" → mysql -u clerk_user -p → Enter: Clerk@456

```
C:\Program Files\MySQL\MySQL Server 8.4\bin>mysql -u clerk_user -p
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.4.7 MySQL Community Server - GPL

Copyright (c) 2000, 2025, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> _
```

```
SELECT * FROM MOVIES;
```

```
mysql> SELECT * FROM MOVIES;
```

MovieID	MovieTitle	Genre	DurationMinutes
1	Avengers: Endgame	Action	181
2	Oppenheimer	Drama	180
3	Barbie	Comedy	114
4	John Wick 4	Action	169
5	Dune: Part Two	Sci-Fi	166
6	Spider-Man: Across the Spider-Verse	Animation	140
7	Godzilla Minus One	Action	125
8	Your Name	Animation	106
9	Suzume	Animation	122
10	Demon Slayer: Mugen Train	Animation	117

```
10 rows in set (0.00 sec)
```

DELETE FROM MOVIES WHERE MovieID = 1; (ERROR)

```
mysql> DELETE FROM MOVIES WHERE MovieID = 1;
ERROR 1142 (42000): DELETE command denied to user 'clerk_user'@'localhost' for table 'movies'
```

-- Log out (using cmd/bash): exit

```
mysql> exit
Bye
```

2. Secure customer information and payment history (if included).

-- Hide phone number

CREATE VIEW CustomerSafeView AS

SELECT

CustomerID,

CustomerName,

CONCAT('****', RIGHT(PhoneNumber, 3)) AS PhoneMasked

FROM CUSTOMERS;

-- Only allow clerk uses VIEW

GRANT SELECT ON CinemaDB.CustomerSafeView TO 'clerk_user'@'localhost';

TEST CASE

SELECT * FROM CUSTOMERS; (ERROR)

```
mysql> SELECT * FROM CUSTOMERS;  
ERROR 1142 (42000): SELECT command denied to user 'clerk_user'@'localhost' for table 'customers'
```

SELECT * FROM CustomerSafeView; (Success)

```
mysql> SELECT * FROM CustomerSafeView;  
+-----+-----+-----+  
| CustomerID | CustomerName | PhoneMasked |  
+-----+-----+-----+  
| 1 | Ava Thomas | ****260 |  
| 2 | James Wilson | ****363 |  
| 3 | Sophia Williams | ****187 |  
| 4 | Isabella Martinez | ****070 |  
| 5 | Ethan Taylor | ****849 |  
| 6 | Michael Anderson | ****695 |  
| 7 | Michael Thomas | ****411 |  
| 8 | Noah Davis | ****255 |  
| 9 | Leo Thomas | ****588 |  
| 10 | Alice Jones | ****900 |  
| 11 | Lucas Martinez | ****481 |  
| 12 | James Williams | ****483 |  
| 13 | Liam Brown | ****321 |  
| 14 | Isabella Martinez | ****010 |  
| 15 | Emily Taylor | ****899 |  
| 16 | Isabella Brown | ****008 |  
| 17 | Alice Martinez | ****995 |  
| 18 | Zoey Johnson | ****003 |  
| 19 | Ethan Rodriguez | ****483 |  
| 20 | Michael Williams | ****744 |
```

3. Set up regular backups and recovery plans.

Backup

```
cd "C:\Program Files\MySQL\MySQL Server 8.4\bin"
```

```
mysqldump -u root -p CinemaDB > C:\Misa\Introduction to Database\backup.sql
```

```
C:\Users\anngu>cd "C:\Program Files\MySQL\MySQL Server 8.4\bin"  
C:\Program Files\MySQL\MySQL Server 8.4\bin>mysqldump -u root -p CinemaDB > "C:\Misa\Introduction to Database\backup.sql"  
Enter password: *****  
C:\Program Files\MySQL\MySQL Server 8.4\bin>_
```

Stimulate data loss

```
DROP TABLE TICKETS;
```

Restore

```
mysql -u root -p CinemaDB < "C:\Misa\Introduction to Database\backup.sql"
```

```
C:\Users\anngu>cd "C:\Program Files\MySQL\MySQL Server 8.4\bin"

C:\Program Files\MySQL\MySQL Server 8.4\bin>mysqldump -u root -p CinemaDB > "C:\Misa\Introduction to Database\backup.sql"
Enter password: *****

C:\Program Files\MySQL\MySQL Server 8.4\bin>mysql -u root -p CinemaDB < "C:\Misa\Introduction to Database\backup.sql"
Enter password: *****

C:\Program Files\MySQL\MySQL Server 8.4\bin>
```

Verification

SELECT * FROM TICKETS;

➔ Table TICKETS reappears

4. Optimize screening lookup and ticket queries for performance.

-- Create index

CREATE INDEX idx_movie_title ON MOVIES(MovieTitle);

CREATE INDEX idx_screening_datetime ON SCREENINGS(ScreeningDate, ScreeningTime);

CREATE INDEX idx_ticket_screening ON TICKETS(ScreeningID);

(If each index is already there, just use: DROP INDEX [name an index] ON [name the table];)

Example: (idx_ticket_screeningTICKETS)

-- Check index

SHOW INDEX FROM MOVIES;

	Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment
►	movies	0	PRIMARY	1	MovieID	A	10	NULL	NULL		BTREE		
	movies	1	idx_movie_title	1	MovieTitle	A	10	NULL	NULL		BTREE		

EXPLAIN SELECT *

FROM MOVIES

WHERE MovieTitle = 'Oppenheimer';

	id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
►	1	SIMPLE	MOVIES	NULL	ref	idx_movie_title	idx_movie_title	802	const	1	100.00	NULL

-- Actual query

SELECT

c.CustomerName,

m.MovieTitle,

```
s.ScreeningTime,  
t.SeatNumber  
FROM TICKETS t  
JOIN CUSTOMERS c ON t.CustomerID = c.CustomerID  
JOIN SCREENINGS s ON t.ScreeningID = s.ScreeningID  
JOIN MOVIES m ON s.MovieID = m.MovieID  
WHERE s.ScreeningDate = '2026-04-10';
```

	CustomerName	MovieTitle	ScreeningTime	SeatNumber
►	Noah Moore	Suzume	13:30	F6
	Ethan Anderson	Suzume	13:30	C5
	Sophia Taylor	Suzume	13:30	F3
	Sophia Davis	Suzume	13:30	D7
	Sarah Taylor	Suzume	13:30	E9
	Lucas Williams	Suzume	13:30	G8
	Alice Anderson	Suzume	13:30	A3
	Mia Thomas	Suzume	13:30	F1